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The Design, Performance, and Field Test Results of a Sourceless Multiphase Flowmeter

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Abstract

The paper discusses recent advancements in phase fraction and flow rate measurement using a microwave-based non-gamma venturi multiphase flowmeter. The primary objectives are to assess measurement performance, system robustness, and usability in various flow loop and field conditions.

The system combines microwave transmission antennas, a Venturi and a Multi-Variable Transmitter (MVT) to measure water-liquid ratio, gas volume fraction and flow rates of each phase together with a microwave reflection antenna to provide measurement of salinity. More than 1,000 points were collected during an initial flow loop test campaign to assess the meter's performance in measuring liquid and gas flow rates, water-liquid ratio, and pressure loss, among other parameters. Several systems were deployed in multiple locations to test operational reliability and robustness of the tool. Field tests involved comparing results with radioactive multiphase flowmeter references, assessing calibration requirements, and evaluating system usability.

The system can accurately determine water properties in varying salinities without requiring in-situ water reference measurement. Additionally, no in-situ oil reference measurement is required to achieve metrological performance within target tolerances for flow with oil of different viscosity or types, across the water-liquid ratio inversion point. The field tests demonstrated mechanical robustness and accurate performance of the system. Measurement uncertainty within acceptance criteria was successfully demonstrated on multiple wells. The system's usability was highlighted by its ease of configuration and maintenance, outperforming the reference multiphase flowmeter. The system's ability to measure water-liquid ratio and gas volume fraction accurately was validated, providing reliable data for oil, water, and gas phase flow rates.

This paper presents a novel approach to multiphase flow measurement using microwave technology, eliminating the need for gamma-ray sources. Such a system offers a safer and more efficient alternative to perform multiphase flow measurements for a wide range of applications with less human intervention.

Introduction

Multiphase flow meters (MPFM) have been available on the market for multiple decades and have a proven track record for its effective use in applications for surface well testing, production monitoring, production optimization, fiscal allocation and custody transfer (Theuveny et al. 2001; Moncada et al. 2023). MPFMs are designed for permanent or mobile topside installation use either offshore or on land. They measure flow rates of oil, water and gas flowing through a pipe in real-time without fluid separation or conditioning. They provide the essential well parameters to monitor the well performance such as liquid flow rate, water-cut (WC) and gas-oil ratio (GOR) for oil wells and gas flow rate, water-gas ratio (WGR) and condensate-gas ratio (CGR) for gas wells.

The application of MPFM's has become increasingly critical, particularly with the rapid deployment of digital solutions for reservoir management and production optimization (Almurayshid et al. 2024). MPFM's play an important role in real-time production monitoring, especially in brownfield developments where artificial lift systems and enhanced oil recovery (EOR) techniques are commonly employed (Tonkonog et al. 2021). The latest success in unconventional field development also showed the importance of real-time production monitoring with MPFM's. Their ability to provide accurate, continuous flow measurements makes them essential for effective field management and decision-making. These emerging trends in reservoir management are driving the need for the deployment and management of a significantly larger number of MPFMs. This shift underscores the importance of advancing MPFM technology to ensure reliable performance with minimal intervention, even as production fluid characteristics evolve over time. This paper provides an overview of existing techniques, their limitations and introduces the new MPFM addressing the modern challenges of production metering.

MPFMs are based on a variety of measurement methods for fraction and velocity measurement. Many MPFMs use differential pressure (DP) transmitters to measure the difference in the pressure at two given points inside the pipe. A venturi meter creates a disturbance to the flow which enables the DP transmitters to measure the pressure difference between the two points. Using Bernoulli's principle, it is known that increasing the velocity of the fluid will cause a decrease in pressure. By obtaining the differential pressure, the total mass flow rate of the fluid can be calculated (Moncada et al. 2023).

Fraction measurements can be achieved with gamma densitometry which employs the use of chemical source, such as barium. Use of dual-energy gamma ray technique with standard source, and lately triple-energy with an addition fluorescence X-rays emitted by the modified source (XRF source) provide accurate measurements of the fractions even for conditions where gas volume fraction (GVF) and water-liquid ratio (WLR) are high. The fluorescence X-rays emitted by the XRF source leads to cutting the WLR uncertainty by more than half for $GVF > 95\%$ when compared with an MPFM equipped with standard source, thus significantly improving the oil flow rate accuracy at levels required for fiscal allocation (Fiore et al. 2024). Despite superior metrological performance, the use of chemical sources is associated with additional safety and environmental regulations. Half-life nature of the chemical source requires it to be changed on a periodic basis depending on remaining activity strength and dispose accordingly resulting in additional maintenance and site visits.

Another challenge with this method is that without additional measurements of salinity, density/composition or in-situ reference measurement to update mass attenuation coefficients, such method is sensitive to salinity changes resulting in underestimation of WLR and consequently incorrect oil, gas and water flow rates. It is especially amplified during flowback and frac plug drill-out operation on coiled tubing to clean out the well, remove proppant left in the horizontal section after the fracturing treatment and plug debris before it is put on production frac flowback operations. During such an operation, salinity can increase from 10 kppm to 140 kppm in less than two days. Under these conditions, setting up an MPFM without the salinity sensor requires the standard procedure of manual collection of water samples to measure the water density or salinity to update the MPFM water calibration points. The procedure is labor-intensive, as

the samples need to be taken at high frequency, e.g., once every hour, to capture the fast changes in water properties (Zhan et al. 2025). Moreover, the EOR application with the water salinity change in the reservoir and production also requires the continuous update of the water mass attenuation to accurately measure the water-cut and oil and water flow rates and assess the sweeping efficiency.

Alternative techniques of fraction measurements using non-radioactive methods include but not limited to microwave transmission and/ or reflection (Karimi et al. 2016; Sharma et al. 2018); electrical impedance (capacitance and conductance/resistance), probe or array, tomography (Mohamad et al. 2016; Jia et al. 2017; Wang et al. 2018); multi-wavelength optical absorption sensor/imaging array (Ünalmis et al. 2016); ultrasonic (Su et al. 2018; Black et al. 2023); nuclear magnetic resonance (Zargar et al. 2021). Although sourceless methods remove the hurdle of handling chemical radioactive sources, their metrological performance has not yet arrived in the same league. This will be further discussed in the paper.

Design and overview of MPFM

Numerous factors can influence the measurement of multiphase flows, and therefore the measurement method should be adapted to the variety of the flow regimes that could be encountered on an oil & gas field and should be able to track rapid production transient phenomena.

In the initial phase of concept identification, 18 different concepts were evaluated, including commercially available meters. Subsequent analysis reduced the number of concepts to 11, and ultimately to a shortlist of 5, based on the assessment provided by the experts group on their capability to meet the defined 13 key requirements grouped in 5 key attributes: performance, cost, range, flow disturbances and sensitivity to fluids.

Further assessment and tests allowed to arrive to final concept selection which is based on microwave antennas array at vertical Venturi throat section. Initial experiments showed that the microwave design meets the liquid and gas rates metering requirement. They also showed a good sensitivity to changes in WLR and GVF observed at high RF operating frequency. Several other prototypes with improved microwave antennas positioning architecture and faster data acquisition rate were prototyped to achieve the required performances.

One of the key advantages of the selected concept, in addition to its performance range and ability to measure salinity, was the reuse of dielectric technology from downhole tools, due to its extensive development history. During 1970-1984, an electromagnetic propagation tool was developed, which uses the travel time and attenuation of microwave-frequency electromagnetic waves to determine the amount of water in rock pores (Serra, 1984). The tool operated in longitudinal and transverse polarization modes at 1.1GHz, corresponding to measurements in horizontal and vertical planes with respect to the tool. Additionally, between 1980 and 1990, a deep electromagnetic propagation tool was developed (Huchital et al., 1982). This device emitted electromagnetic energy into the formation surrounding the well bore at 25 MHz. Measurements of the attenuation and velocity (wavelength) of this electromagnetic wave provided the resistivity and dielectric constant of the formation. The deep propagation tool used a unique antenna array to return electrical parameters characteristic of the uninvaded formation. Lastly, during 2002-2012, an array dielectric tool was developed (Donadille et al., 2016). It offered high metrological performance in complex environments and is still widely used for reservoir characterization. The expertise in dielectric measurements accumulated over decades for downhole applications was effectively transferred to surface MPFM measurements.

The new sourceless MPFM addresses the challenge of multiphase flows measurements by using high sampling-rate, multi-view, multi-frequency microwave measurements to capture multiphase-flow dynamics in time and space, based on the variation in fluid physical properties (complex permittivity), to precisely determine the individual phase fractions. In this manner, the MPFM measurements are unaffected by any flow pattern and salinity change that appear in multiphase flow lines. The MPFM is based on technologies

that combine a differential pressure measured across a Venturi tube, with a microwave sensor that measures the individual phase fractions inside the Venturi.

MPFM is comprised of the microwave acquisition module for phase fraction measurement, and the Venturi with MVT for total mass flow rate determination. Microwave acquisition module consists of microwave transmission for phase fraction measurement and microwave reflection for salinity measurement. The design also uses an isolation block that allows easy isolation of the MVT when it needs to be shipped for calibration, or when performing a zero trim on the transmitters in place (without removing the MVT from the Venturi). It is a good option for calibration checks in the field, or if a MVT needs replacing as there is no need to bypass the meter. Additional MVT performance is achieved by remote seals which allow pressure measurement to be made directly on the venturi, reducing the need for impulse tubing.

Microwave transmission measurement is performed by multi-plane magnetic-dipole dielectric-window aperture antennas flush-mounted at the Venturi throat section. The bulk flow velocity of a multiphase flow at Venturi throat can be measured via cross-correlation transit-time processing of dual-plane transmission signals at high data sampling rate. With the Venturi DP as an input, gas volume fraction (GVF) can be derived.

Microwave reflection is performed by microwave reflection probe flush-mounted at the Venturi inlet section (Zhan et al. 2022). The microwave reflection probe measures the permittivity and conductivity of the near-wall liquid-rich region for water salinity determination for a wide range of GVF. The real-time salinity measurement by microwave reflection allows phase fraction measurement by microwave transmission to be independent of salinity change of the flows. The bulk mixture permittivity and conductivity can also be derived from multi-frequency microwave transmission signals of amplitude-attenuation and phase-shift via physics-based models (Xie et al. 2023). Water/liquid ratio (WLR) can be interpreted from mixture permittivity and conductivity via gas-liquid dielectric mixing models with GVF and real-time salinity as an input. The MVT measures Venturi DP, line temperature (T) and line pressure (P) for a multiphase flow. In-line mixture density can be corrected by real-time salinity measurement and embedded Pressure, Volume, Temperature (PVT) model. With the Venturi flow model, the total mass flow rate can be derived from DP, Venturi geometry parameters and in-line mixture density after accounting for energy losses. The individual phase flow rates are calculated from the total mass flow rate, GVF and WLR measured (Figure 2). The use of physics-based modelling approach allowed to produce consistent results in a variety of multiphase flow conditions and exclude interpretation model variability.

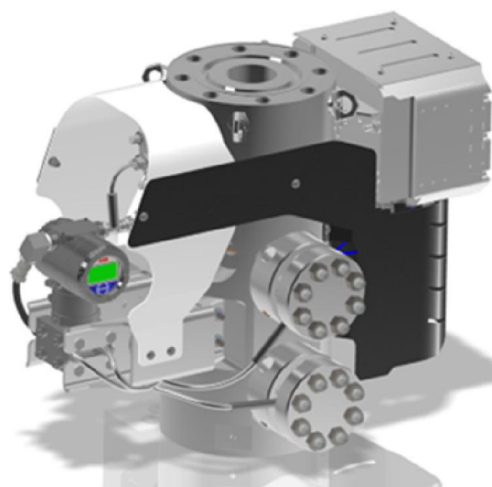


Figure 1—General view of the MPFM.

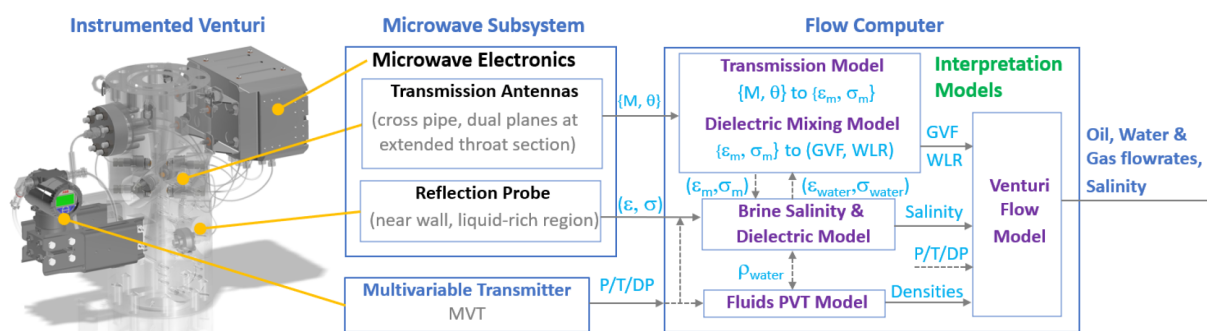


Figure 2—Microwave transmission and reflection measurements.

Nonetheless, further improvements in interpretation models such as use of hybrid machine learning and physics-based modelling are feasible and being explored.

Test Description

Laboratory tests

Although not discussed in great details in this paper, the MPFM undertook a rigorous testing to validate reliability and performance for surface well testing mission profile and use in hazardous environment. The tests included but were not limited to accelerated lifetime, humidity, storage and operating temperature, operating and non-operating vibration, and shock tests at various component, sub-assembly and final assembly levels.

Flow loop tests

Flow loop tests described in this paper were conducted in several test facilities. Most of test data was collected in an ISO-17025 accredited flow loop test facility located in Singapore. The equipment and piping were organized into 7 separate sections corresponding to the process P&IDs, as shown in Figure 3.

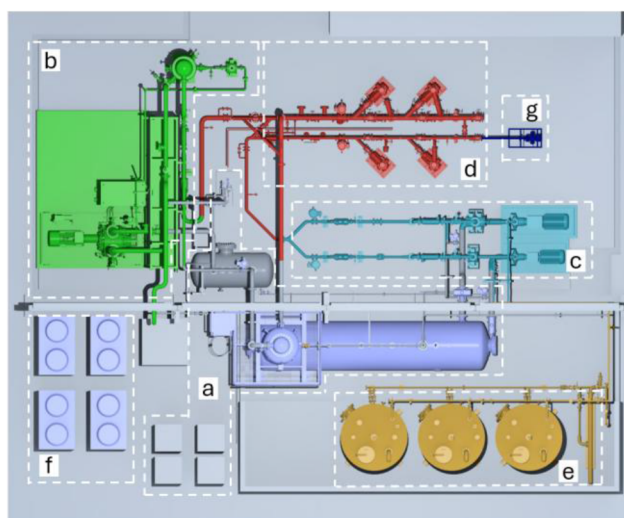


Figure 3—General layout of the flow loop: (a) Separation system; (b) Compressors section; (c) Pumps section; (d) Test section; (e) Storage and transfer; (f) Cooling system; (g) Device under test (e.g MPFM).

The gas, oil, and water phases were selected to test the meters under safe, reproducible, and realistic conditions while also facilitating the three-phase separation in the separator. The flow loop was filled and pressurized with high-purity nitrogen (99.99%). The oil used for the test campaign was Exxsol D80.

Throughout most of the operating range of the flow loop, the combined uncertainties on mass flow rates were: less than 0.81% for gas, less than 0.59% for oil, and less than 0.42% for water. The MPFM was connected to the test station of a 3-inch test line (Figure 4).



Figure 4—Illustration of the MPFM setup during flow loop tests.

The test matrix consisted of flow periods that are within the working envelope of the flow loop instrumentation. The flow periods were selected to cover a wide range of flow situations in a short period of time, details of selected flow periods are illustrated on Figure 5 which shows distribution of flow periods as a function of liquid and gas volumetric flow rates, and Figure 6.

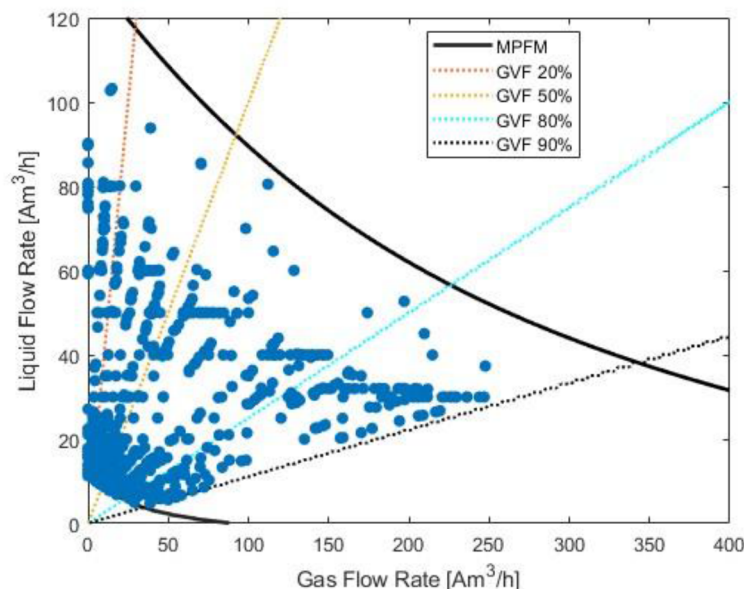


Figure 5—Flow periods distribution as function of liquid and gas volumetric flow rates.

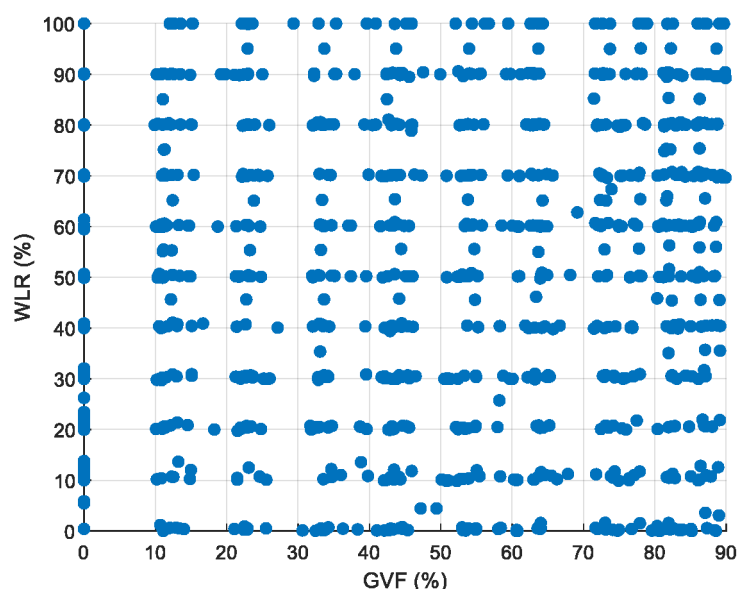


Figure 6—Flow periods distribution as function of WLR and GVF.

The flow loop aqueous phase had a varying concentration of NaCl from 18 kppm to 90 kppm whereas line pressure conditions varied from 7 to 31 bara.

Field tests

The sourceless field test units were tested in series with an MPFM equipped with standard source in order to provide field-condition references of WLR, GVF and flow rates. They had the following specifications:

- MPFM Interface, Pressure and Temperature specification:
 - Flange interface: ASME Flange, 3 inch, CL1500.
 - Process Pressure: 3,000 psig / 207 barg at 121 degC.
- Skid Mounted MPFM Interface, Pressure and Temperature specification:
 - Flange interface: ASME Flange, 3 inch, CL600 / 3 inch Weco Figure 602.
 - Process Pressure: 1,440 psig / 99.3 barg at 38 degC and 1,333 psig / 92 barg at 121degC.
- Process Temperature: -20 degC to 121 degC.
- Venturi profile: inlet ID d=74mm, throat ID d=37mm ($\beta=0.5$).
- Three microwave transmitter-receiver planes.
- Microwave reflection probe: at upstream of venturi convergent.
- MVT (DP/p/T): remote seals and isolation blocks.
- Material: Duplex, Sour Service in accordance NACE MR0175/ISO15156; Inconel, full Sour Service in accordance NACE MR0175/ISO15156.
- Temperature thermowell/PT100 probe at downstream Venturi.
- Intrinsically safe microwave-transmission and reflection measurement electronics.
- Flow computer.

The single leg unit of sourceless MPFM was connected (at the upstream) to a standard source MPFM in series as per the schematic shown in Figure 7.

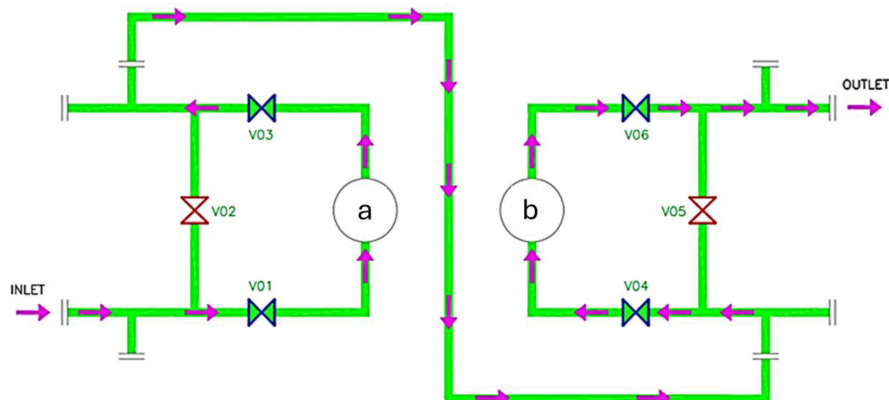


Figure 7—Schematic of MPFM's connection: (a) Sourceless MPFM, (b) Standard source MPFM.

Consequently, the field test unit was installed on a mobile testing skid for application in mobile well testing operations (Figure 8). This approach allowed the field test units to be subjected to various well production conditions throughout the duration of the field tests.



Figure 8—Sourceless MPFM (right) setup in series with standard source MPFM (left) on a trailer.

The environmental stress exposure was expected also to be higher than for regular mission profile for permanent installation and therefore it would help to identify potential weak points in the design. Field test units were deployed in Asia, Middle East and North Africa. The field test aimed to evaluate:

- Measurement repeatability, drift, and calibration needs with reference measurements and empty pipes.
- Regular empty-pipe and brine measurements, ensuring raw and borehole-compensated data stability.
- Calibration frequency for oil permittivity and water conductivity from full-oil and full-brine readings using a calibrated reflection probe with p/T modeling.
- System robustness, reliability, and operability by checking electronics or mechanical hardware failures, required maintenance, and usability issues.
- Software capability for configuring the system, plotting data, and exporting data including electronics hardware frequency/gain settings and diagnostic values like board temperature.

- Usability improvement through staged approach of interpretation models development:
 - Enhance the interpretation model (WLR/GVF/flowrate) based on flow loop tests.
 - Phase 1: Add MPFM-reference field test data to extend the model.
 - Phase 2: Test the new model's performance on other field test wells.
 - Compare extended model flowrates with reference MPFM results for accuracy.

Success criteria include acceptable metrological performance versus reference, effective system functionalities/usability, and stable, robust microwave sensor response with in-situ measurement.

Results and Discussion

In the processing workflow of the MPFM, the effect of salinity is embedded in the information of water conductivity and water permittivity that are used in the dielectric mixing model to derive WLR from mixture conductivity and mixture permittivity measured by microwave transmission via the transmission model.

To obtain water conductivity and water permittivity in the case of changing salinity, there are three methods:

1. Perform full water point (water in-situ process), obtain water permittivity and water conductivity from transmission model.
2. Use measured water density at given line pressure and temperature through dielectric model (also known as salinity & brine dielectric models) to obtain salinity. Calculate water permittivity and water conductivity from salinity.
3. For version with microwave reflection probe that measures real-time salinity at given line pressure and temperature, calculate water permittivity and water conductivity from reflection-output salinity.

The results obtained from the 3 methods are evaluated in the 4 sets of data with a range of salinity from 25 kppm to 90 kppm. For method 1, water properties are derived from full water point via transmission model in the flow loop. For method 2, the water density is measured by Coriolis meter in the flow loop at the line temperature and pressure condition. For method 3, after data quality control to ensure measurement condition is water-continuous, salinity can be measured by microwave reflection probe at the Venturi inlet.

The comparison of water permittivity and conductivity evaluated by the three methods described is shown in Figure 9.

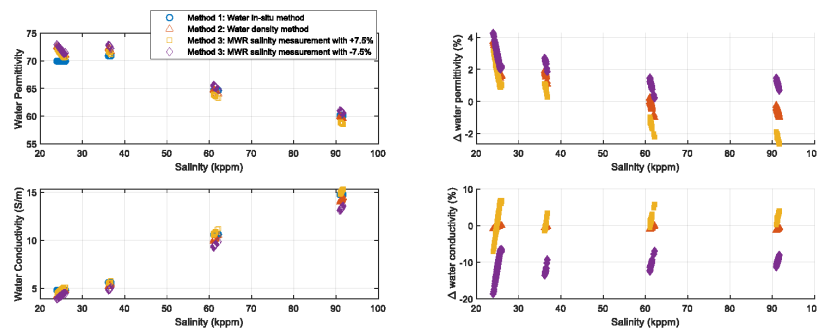


Figure 9—Comparison of water permittivity, and water conductivity between 3 methods: method 1 (water in-situ), method 2 (water density) and method 3 (reflection probe measurement; simulate with a $\pm 7.5\%$ measurement in salinity).

To understand how the difference in water conductivity and permittivity obtained by using the 3 methods impact the metrological performance, further analysis of deviation in performance of WLR, liquid rate and gas rate was performed. It was observed that the performance using method 1 (water in-situ) to obtain water

properties is similar to that using method 2 (water density), with the difference in performance in WLR, liquid rate and gas rate less than 1%. For method 3 of using reflection probe to measure real-time salinity, and update water permittivity water and conductivity, it is observed that the performance using method 1 (water in-situ) to obtain water properties is similar to that using method 3 (reflection probe measurement; simulate with a $\pm 7.5\%$ measurement in salinity), with the difference in performance in WLR, liquid rate and gas rate less than 1%.

Further on, sensitivity of the processing workflow to the oil viscosity, which would cause a shift in inversion point of WLR, was evaluated using two types of the oil - Exxol D80 and Exxol D120. The oil permittivity is 2.1 by default. The oil viscosity of Exxol D120 was 4.2 cP at line temperature 20 degC, which is 3 times that of oil viscosity 1.4 cP of Exxol D80 at a line temperature of 30 degC. Despite both being light oil, there is still difference in the WLR inversion point as shown in Figure 10.

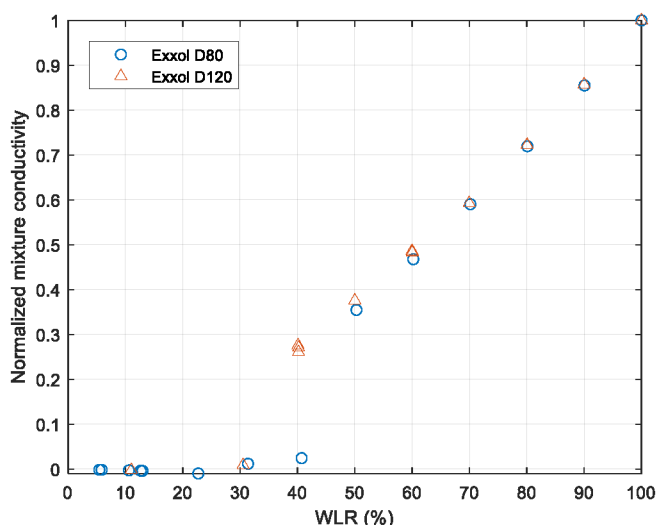


Figure 10—Variation of normalized mixture conductivity (by water conductivity) against WLR for oil-water liquid flows.

WLR inversion point, or phase inversion point is the WLR where the flow transits from oil continuous region to water continuous region. The transition is marked by a significant increase in the conductivity of the flow, as water phase becomes the dominant phase. From Figure 10, it is observed that the phase inversion point occurs at a higher WLR of 40% ~ 50% for Exxol D80 than Exxol D120 which sees a markable rise in normalized mixture conductivity at a WLR of 30% ~ 40%.

As the oil-continuous zone and water-continuous zone are differentiated by conductivity instead of WLR value in the processing workflow, the workflow is robust to differentiate the oil-continuous and water-continuous zones, despite the different point of phase inversion resulted from different oil viscosities. This was evident in the metrological performance for all studied data sets within the metrological acceptance criteria.

To demonstrate the significance of real-time salinity tracking for accurate measurement of MPFMs, a simulation over 15 years of production is performed. In the simulation, oil production and salinity follow an exponential decay from 2nd year to the 15th year. WLR follows an exponential increase from 2nd year to the 15th year. The results in salinity, oil flow rate and WLR are assessed with relative salinity measurement accuracy of 7.5% and without salinity tracking. The details of the simulation setup are shown in Table 1.

Table 1—Simulation configuration.

Simulation parameters	Initial salinity (kppm)	15 th year salinity (kppm)	Initial Qo (bpd)	15 th year Qo (bpd)	Initial WLR (%)	15 th year WLR (%)	T (degC)	P (bar)	Total (true) Qo (bbl)
	70	35	2,000	500	50	90	60	30	5,869,055

The simulation results are shown in Figure 11 and Table 2.

From Figure 11 and Table 2, it is observed that without real-time salinity tracking, WLR tends to be overpredicted over time with decreasing salinity, resulting in an underestimation of oil production. In the simulated case, the total oil rate produced measured without salinity tracking is underpredicted by 8.59% compared to the true value. With real-time salinity tracking, the continuous update of water properties ensures accurate WLR and Qo measurement.

Table 2—Qo simulated over 15 years with relative salinity measurement accuracy of 7.5% and without salinity tracking.

	Total (true) Qo	Qo with +7.5% relative salinity measurement accuracy	Qo with -7.5% relative salinity measurement accuracy	Qo without salinity tracking
Total Qo	5,869,055	5,772,413	5,965,901	5,364,722
Deviation compared to total (true) Qo	-	-1.65	1.65	-8.59

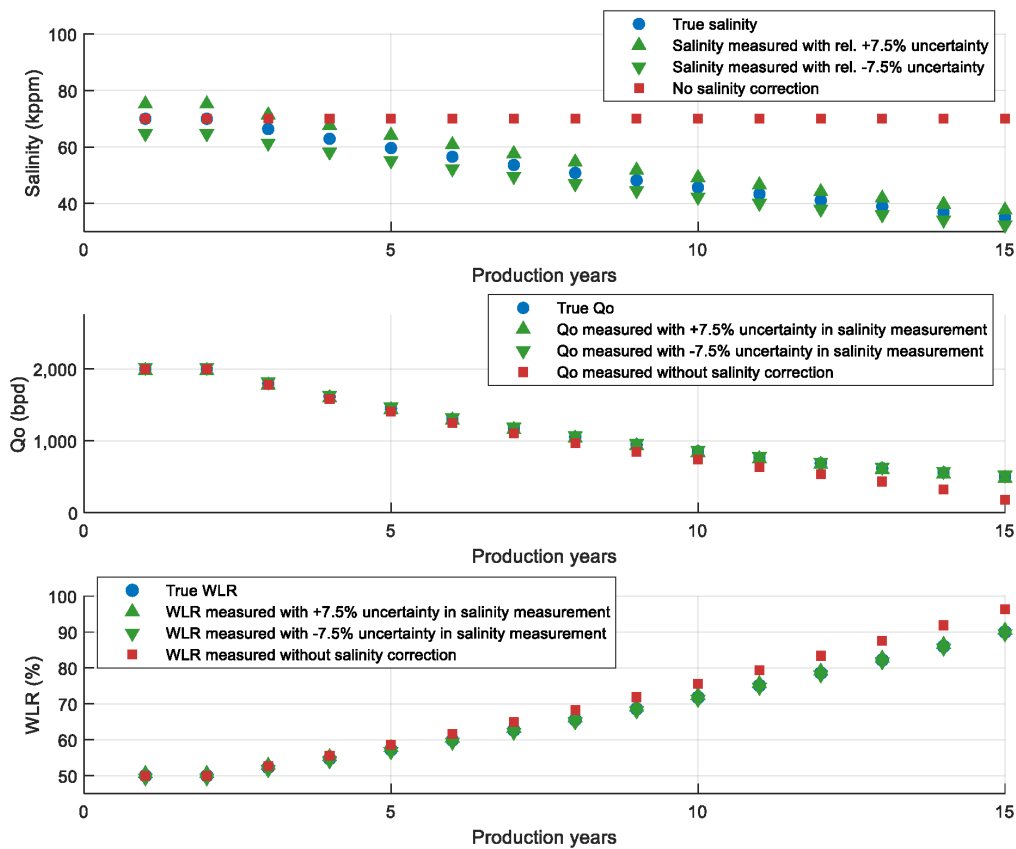


Figure 11—Simulation results on salinity, oil flow rate (Qo), and WLR over 15 years of production with relative salinity measurement accuracy of 7.5% and without salinity tracking are evaluated against true values defined in the simulation.

The new MPFM was designed for permanent metering in fixed installations. To test the prototype under stress, a mobile field test program was implemented to ensure it operated well in harsh conditions. Compared to permanent setups, the field test exposed the system to higher cyclic stress from process flow (pressure and temperature), increased shock and vibration from trailer transport, greater shock and vibration from piping setup, elevated corrosion due to repeated exposure to process fluid and ambient air (depressurization).

Particular attention was paid to the condition of the unit over the whole field test period. The units were inspected for defects (abnormal aging or corrosion, bolt loosened, parts failure) and the stability of the air point was verified during the period. Regular physical inspection verified mechanical robustness and corrosion resistance.

The field test campaign was mainly focused on the metrology consistency against a trusted reference flowmeter connected in series. During this field test, the system calibration and measurement stability was checked. The first phase of the field test included the data from production testing under stable flow conditions for many wells with large range of GVF from 10 to 94 % and water-cut from 0 to 96 %. During the second phase the validation of measurement stability on a long term (a minimum of 1 month) well test on a stable well was performed. The meter was tested during the oil well flowback following acid stimulation in the third phase of operations. This phase was characterized by changes in water salinity during production. The MPFM was utilized to digitize the flowback process by providing automated water-cut measurements without the need for manual calibration.

All field test objectives have been achieved (Table 3). The mechanical robustness and corrosion resistance have been verified based on the field test exposure. Measurement uncertainty of well tests was within acceptance criteria and was successfully demonstrated on multiple wells. System and software usability were assessed during both field tests and end user tests. They highlighted the ability to configure easily the system and monitor results and export data which is achieved more efficiently when compared to reference MPFM as in-situ is not required and there is no handling of radioactive source.

Table 3—Field test objectives.

Objective description	Target	Status
Measurement stability	Raw-data (Amp, Phase)(± 0.1 dB, $\pm 1^\circ$) Borehole-compensated data (± 0.25 dB, $\pm 0.25^\circ$)	
Metrology performance consistency vs reference MPFM	Range WLR 0-100%, GVF 20%- 90%	
Robustness to land operation (rig-up and transportation)	No hardware failure	
Software	Confirm ability to configure easily the system and monitor results and export data	
Field calibration robustness		
Operation start-up time	≤ 1 hour	
Maximum time to repair	1 day	
Assess “plug and play” / easiness to install / operate / calibrate	Confirm ability to configure easily the system	

During the field test, an analysis of the total cost of ownership (TCO) was conducted. This analysis included an evaluation of acquisition costs, operating costs, scheduled maintenance, and unscheduled maintenance costs over the equipment's lifetime. The study compared a reference multiphase flow meter (MPFM) with the standard source to a sourceless MPFM with the salinity measurement option. As illustrated

in Figure 12 below, the sourceless MPFM with the salinity measurement option demonstrated nearly 30% reduction in TCO over the equipment's lifetime. This reduction is attributable to lower acquisition costs, as there is no chemical source, and decreased maintenance requirements because periodic in-situ reference measurements are not necessary to update mass attenuation coefficients due to changes in water properties.

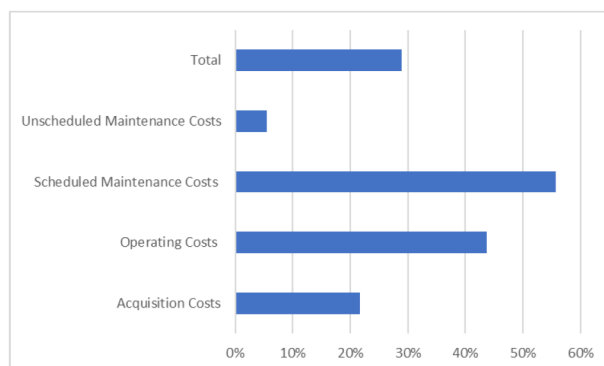


Figure 12—TCO reduction by using sourceless MPFM under study vs standard source MPFM.

Sourceless MPFM had a narrower operating range for GVF compared to standard source and XRF MPFMs. While its metrological performance met acceptance criteria, it was still less accurate than radioactive-based MPFMs.

Conclusions

The sourceless MPFM accurately determined water properties in varying salinities without requiring in-situ water reference measurement. No in-situ oil reference measurement was required to achieve metrological performance within target for oil of different viscosities across the water-liquid ratio inversion point. The field tests demonstrated mechanical robustness and accurate performance of the system. Measurement uncertainty within acceptance criteria was successfully demonstrated on multiple wells. The system's usability was highlighted by its ease of configuration and maintenance, outperforming the reference MPFM. The system's ability to measure water-liquid ratio and gas volume fraction accurately was validated, providing reliable data for oil, water, and gas phase flow rates.

Radioactive-based MPFM is best if metrology and operating envelope are driving factors. Microwave type MPFM leads to lower TCO over equipment lifetime as there is no need to handle radioactive sources and conduct periodic in-situ reference measurements due to changing water properties. In microwave type MPFM the concept of analytical model was pushed to extreme. This is best that can be achieved with physics-based model meter, anything better is a result of empirical based tuning to flow loop and field test data. Potentially with data analytics it can be enhanced.

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